

CVAT Overview

The open-source tool for image and video annotation

Machine learning systems often struggle due to poor-quality data. Without effective tools, improving a model can be tough and inefficient.

CVAT.ai is a versatile **tool** for **annotating images and videos**, serving the computer vision community worldwide.

Our goal is to help developers, businesses, and organizations globally by using a Data-centric AI approach.

CVAT offers three versions:

- **CVAT Cloud:** Start online with CVAT, **available for free**. You can also choose a **subscription** for **unlimited data, collaboration, auto-annotations**, and **more**.
- **Self-hosted CVAT Community Edition:** Follow the **Self-hosted Installation Guide** for setup.
- **Self-hosted CVAT Enterprise Edition:** We provide **Enterprise-level support** for this version, **including premium features** like **SSO, LDAP**, advanced integrations with **Roboflow and HuggingFace**, and **advanced analytics**. We also offer **professional training** and **24-hour SLA support**.

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Tools and formats

CVAT stands as a comprehensive tool for image and video annotation, essential for various computer vision tasks.

It emphasizes user-friendliness, adaptability, and compatibility with a range of formats and tools.

Supported formats

CVAT’s supports the following formats:

- **For 3D:** `.pcd` , `.bin`
- **For image:** everything supported by the Python **Pillow library**, including formats like `JPEG` , `PNG` , `BMP` , `GIF` , `PPM` and `TIFF` .
- **For video:** all formats, supported by ffmpeg, including `MP4` , `AVI` , and `MOV` .

For annotation export and import formats, see [Export annotations and data from CVAT](#)

Annotation tools

CVAT offers a wide range of annotation tools, each catering to different aspects of image and video labeling:

Annotation Tool	Use Cases
3D Object Annotation	Ideal for projects that require depth perception and volume estimation, like autonomous vehicle training.
Attribute Annotation Mode	Useful for adding detailed information to objects, like color, size, or other specific characteristics.

Annotation with Rectangles	Best for simple object detection where objects have a box-like shape, such as detecting windows in a building.
Annotation with Polygons	Suited for complex shapes in images, like outlining geographical features in maps or detailed product shapes.
Annotation with Polylines	Great for annotating linear objects like roads, pathways, or limbs in pose estimation.
Annotation with Ellipses	Ideal for objects like plates, balls, or eyes, where a circular or oval annotation is needed.
Annotation with Cuboids	Useful for 3D objects in 2D images, like boxes or furniture in room layouts.
Annotation with Skeletons	Ideal for human pose estimation, animation, and movement analysis in sports or medical fields.
Annotation with Brush Tool	Perfect for intricate and detailed annotations where precision is key, such as in medical imaging.
Annotation with Tags	Useful for image and video classification tasks, like identifying scenes or themes in a dataset.

These tools make CVAT a versatile platform for a range of annotation needs, from basic labeling to complex, multidimensional tasks in advanced computer vision projects.

Automated labeling

CVAT has an automated labeling features, enhancing the annotation process significantly, potentially speeding it up by up to 10 times.

 **Note:** For more information, see [OpenCV and AI Tools](#)

Below is a detailed table of the supported algorithms and the platforms they operate on:

Algorithm Name	Category	Framework	CPU Support	GPU Support
Segment Anything	Interactor	PyTorch	✓	✓
Deep Extreme Cut	Interactor	OpenVINO	✓	
Faster RCNN	Detector	OpenVINO	✓	
Mask RCNN	Detector	OpenVINO	✓	
YOLO v3	Detector	OpenVINO	✓	
YOLO v7	Detector	ONNX	✓	✓
Object Reidentification	ReID	OpenVINO	✓	
Semantic Segmentation for ADAS	Detector	OpenVINO	✓	
Text Detection v4	Detector	OpenVINO	✓	
SiamMask	Tracker	PyTorch	✓	✓
TransT	Tracker	PyTorch	✓	✓
f-BRS	Interactor	PyTorch	✓	
HRNet	Interactor	PyTorch		✓
Inside-Outside Guidance	Interactor	PyTorch	✓	
Faster RCNN	Detector	TensorFlow	✓	✓
RetinaNet	Detector	PyTorch	✓	✓
Face Detection	Detector	OpenVINO	✓	

Useful links

Name	Description
User Manual	This comprehensive guide covers all CVAT tools available for work. It includes descriptions of all available tools, quality control methods, and procedures for importing and exporting data. This manual is relevant for both CVAT Cloud and Self-Hosted versions.
CVAT Complete Workflow Guide for Organizations	This guide provides a comprehensive overview of using CVAT for collaboration in organizations.
Subscription Management	Learn how to choose a plan , subscribe, and manage your subscription effectively.
XML Annotation Format	Detailed documentation on the XML format used for annotations in CVAT essential for understanding data structure and compatibility.

Self-Hosted

Name	Description
Self-hosted Installation Guide	Start here to install self-hosted solution on your premises.
Dataset Management Framework	Specifically for the Self-Hosted version, this framework and CLI tool are essential for building, transforming, and analyzing datasets.
Server API	The CVAT server offers a HTTP REST API for interactions. This section explains how client applications, whether they are command line tools, browsers, or scripts, interact with CVAT through HTTP requests and responses.
Python SDK	The CVAT SDK is a Python library providing access to server interactions and additional functionalities like data validation and serialization.
Command Line Tool	This tool offers a straightforward command line interface for managing CVAT tasks. Currently featuring basic functionalities, it has the potential to develop into a more advanced administration tool for CVAT.
XML Annotation Format	Detailed documentation on the XML format used for annotations in CVAT essential for understanding data structure and compatibility.
AWS Deployment Guide	A step-by-step guide for deploying CVAT on Amazon Web Services, covering all necessary procedures and tips.
Frequently Asked Questions	This section addresses common queries and provides helpful answers and insights about using CVAT.

Integrations

CVAT is a global tool, trusted and utilized by teams worldwide. Below is a list of key companies that contribute significantly to our product support or are an integral part of our ecosystem.

Note: If you're using CVAT, we'd love to hear from you at contact@cvat.ai.

Integrated Service	Available In	Description
Human Protocol	Cloud and Self-hosted	Incorporates CVAT to augment annotation services within the Human Protocol framework, enhancing its capabilities in data labeling.
FiftyOne	Cloud and Self-hosted	An open-source tool for dataset management and model analysis in computer vision, FiftyOne is closely integrated with CVAT to enhance annotation capabilities and label refinement.
Hugging Face & Roboflow	Cloud	In CVAT Cloud, models from Hugging Face and Roboflow can be added to enhance computer vision tasks. For more information, see Integration with Hugging Face and Roboflow

CVAT includes the following licenses:

License Type	Applicable To	Description
MIT License	Self-hosted	This code is distributed under the MIT License, a permissive free software license that allows for broad use, modification, and distribution.
LGPL License (FFmpeg)	Cloud and Self-hosted	Incorporates LGPL-licensed components from the FFmpeg project. Users should verify if their use of FFmpeg requires additional licenses. CVAT.ai Corporation does not provide these licenses and is not liable for any related licensing fees.
Commercial License	Self-hosted Enterprise	For commercial use of the Enterprise solution of CVAT, a separate commercial license is applicable. This is tailored for businesses and commercial entities.
Terms of Use	Cloud and Self-hosted	Outlines the terms of use and confidential information handling for CVAT. Important for understanding the legal framework of using the platform.
Privacy Policy	Cloud	Our Privacy Policy governs your visit to https://cvat.ai and your use of https://app.cvat.ai , and explains how we collect, safeguard and disclose information that results from your use of our Service.

Get in touch

To get in touch, use one of the following channels:

Support Channel	Applicable To	Description
Discord Channel	Cloud and Self-hosted	A space for broader discussions, questions, and all things related to CVAT.
LinkedIn	Cloud and Self-hosted	Follow for company updates, news, and employment opportunities.
YouTube Channel	Cloud and Self-hosted	Find tutorials and screencasts about CVAT tools.
GitHub Issues	Cloud and Self-hosted	Report bugs or contribute to the ongoing development of CVAT.
Customer Support Channel	Cloud (Paid Users)	Exclusive support for CVAT.ai cloud paid users.
Commercial Support Inquiries	Cloud and Self-hosted	For direct commercial support inquiries, email contact@cvat.ai .